AYUSHI AGRAWAL

Researcher & Passionate Educator

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EXPERIENCE

Doctoral Researcher

Cognitive Science Lab, IIIT-H
April 2020 - Present — Gachibowli, Hyderabad

- Engaged in research within the field of Social Neuroscience in the context of Human Empathy structures to promote empathy and combat societal biases.
- Researching the domain of Rape Empathy by investigating behavioral, clinical and neural networks associated with social empathy/cognitive control biased by social perception of women rape victims.
- Utilizing a multidisciplinary approach integrating Eye-tracking technology (Tobii X2-30, Tobii X-120, SR research), Galvanic skin response measurements (Empatica E4, Thought Technology BioInfiniti), fMRI, and standard questionnaires.
- Assisted in EEG data collection for a study investigating upper limb motor tasks, gaining proficiency in operating the BrainVision 32 electrode system.
- Contributed to data acquisition for spine biomechanics research utilizing marker-based motion tracking systems, specifically Opti-track technology.

Teaching Assistant

IIIT-H
2014 - 2016, 2022 - 2024 — Gachibowli, Hyderabad

- Conducted tutorials and facilitated classes, designed assessment questions for a batch of 100 students in both Cognitive Neuroscience (2022, 2023) and Introduction to Brain and Cognition (2024) courses.
- Mentored students in IASNLP workshop 2015 followed by facilitating two-month hands on sessions on 'Word Sense Disambiguation (WSD)' & 'Word-phrase alignment' for 50 NLP interns in 2016.

Assistant Professor

CS Department, S.G.S.I.T.S
2016-2017 — Indore, Madhya Pradesh

- Supervised computer networking and programming labs for C and C++ programming languages for fourth-year and first-year B.Tech students .
- Taught courses like Data Structures, C Programming, and Systems Programming, instructing batches of 100 and 150 students.
- Designed assignments, conducted tutorials, and administered vivas to assess student comprehension and progress followed by marking and grading.

Research Intern

LTRC (Anusaaraka Lab), IIIT-H
2014-2016 — Gachibowli, Hyderabad

- Worked on developing and implementing an algorithm for mutual bootstrapping of the exisiting limited data (linguistic NLP resources).
- Hands on with multiple NLP tools like tokenizer, morph analyser, parsers, dictionaries etc.in course of doing word-phrase alignment using dependency relations, translations from NMTs, SMT tools etc.

OBJECTIVE

"Passionate Social Neuroscience researcher dedicated to unraveling empathy networks shaped by societal biases and stereotypes. Leveraging interdisciplinary skills in neuroimaging, cognitive neuroscience, and social sciences to uncover new insights into social behavior's neural mechanisms. Collaborative research enthusiast committed to continual learning and skill development to drive positive societal change and organizational success through innovative research endeavors."

EDUCATION

PhD (CS), CGPA - 9.5
IIIT Hyderabad — 2020 - Present

PGSSP, CGPA - 8.5
IIIT Hyderabad — 2018 - 2019

M.Tech (CS), CGPA - 8.1
Banasthali University
2013 - 2015

B.Tech (CS), CGPA - 7.3
GGITS — 2009 - 2013

Higher Secondary, CGPA - 7.3
Christ Church Girls School
2008 - 2009

Secondary, CGPA - 8.6
Christ Church School
2006 - 2007

SKILLS

Python | CLIPS | C++ | JAVA | SQL | MYSQL | Oracle 9i | MATLAB | E-Prime | Experiment Builder | Psytoolkit

OPERATING SYSTEMS

Linux | MacOS | Windows

AREAS OF INTEREST

- Social Neuroscience
- Cognitive Neuroscience
- Empathy Networks
- Natural Language Processing
- Machine Learning
- Information Retrieval

PROJECTS

☑ Sexism detection in Movie dialogues

- Developed a model to detect gender bias in movie scripts across different genres and time by utilizing ML, NLP, and sentiment analysis, addressing challenges like automatic character gender identification, analysis of dialogue length, gender distribution across dialogues, gender distribution of the characters, speaking roles etc.
- Utilized IMDbPy library and character's first name features (vowel endings, vowel/consonant counts, ASCII values, etc.) to predict character gender labels, then analyzed dialogue gender differences using semantic and structural features of the dialogues.

☑ English Language Speech Tool - testing for efficacy

- Condensed the voisTutor corpus from 1024 words to 250 words (125 pairs) by leveraging distinctive phoneme features crucial for learning English pronunciation .
- Developed an assistive tool encompassing 94 relevant phonetic patterns to generate randomized 250-word subsets, facilitating effective participant pronunciation training and assessment.

☑ Ensemble learning based module for multiple Parsers and implementation of a WSD inference engine

- Created a post processing module for multiple parser outputs to classify, combine and generate better outputs. Analyzed Stanford dependencies vs. HPSG logon EDS (elementary semantic dependencies) to extract performance patterns for integration of ERG Framework in Anusaaraka.
- Improved WSD inference engine by analyzing linguistic (dependency parsing) and semantic (coarse semantic classification) information present in the sentence, encoding it downstream as inference rules for disambiguation.
- Tools used: Phrase-based SMT system: Stanford Phrasal, Language Accessor cum RBMT system: Anusaaraka (It uses many open source tools like Apertium, Parsers, Morph analyzer, dictionaries etc. in itself), CLIPS.

☑ Phrasal alignment of sentence-aligned corpora

- Created an alignment module using information from linguistic resources, Stanford Parser (English), Dependency parser (Hindi), etc as facts to create a phrase-aligned corpora from sentence aligned parallel corpora.
- Implemented a clause boundary detection and alignment tool based on Stanford Constituency Parser for English and Tense Aspect Modality (TAM) grouping for Hindi. This significantly reduced the wrong inter-clause and intra-clause alignments of phrases.
- Tools used: Rule Based Machine Translation system: Anusaaraka, Apertium, Parsers, Morph analyzer, Stanford Phrasal, CLIPS, Python

☑ Improving Fluency of Anusaaraka using Statistical Machine Translation

- Developed a WSD ranking module using KenLM (Kneser-Ney Language Model) to score, aid and improve Anusaaraka's translations leading to improvement in overall fluency of translations.
- Tools used: Python, Shell script, KenLM (Kneser-Ney Language Model), Anusaaraka

AREAS OF EXPERTISE

- Research Methodology
- Experimental Design
- Data Analysis and Statistics
- Neuroimaging (fMRI, EEG)
- Psychophysiology (GSR)
- Knowledge-based system
- Multimodal Data Integration

PROFESSIONAL REFERENCES

Kavita Vemuri, Ph.D
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IIIT Hyderabad, Gachibowli 500032, India

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PUBLICATIONS

- Objectifying Gaze — An empirical study with Non-sexualised images (CogSci 2024)
- Comparing task-based and resting-state brain networks: A graph learning approach for naturalistic stimuli fMRI (EUSIPCO 2024, under review)
- S. GRS, A. Agrawal et al., "Graph learning methods to extract empathy supporting regions in a naturalistic stimuli fMRI," 2024 [https://arxiv.org/abs/2403.07089].
- Sharma, H., Karan, S. S., Agrawal, A. K., & Vemuri, K. (2023). The role of individual physical body measurements and activity on spine kinematics during flexion, lateral bending and twist tasks in healthy young adults–Comparing marker (less) data. Biomedical Signal Processing and Control, 82, 104517.